OP-CODE: Your Personalized Student Life Manager !!

CS2011: Foundations of Computer Systems

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Welcome to the *"OP-CODE: Your Personalized Student Life Manager"* project !! In this assignment, you will create a comprehensive student life management system in MIPS assembly language. In this assignment you're building a tool that you can actually use in your daily campus life to manage your schedule, track academic progress, and optimize your time. Here, every student's program will be completely personalized based on their student ID and actual academic data. No two submissions should be same !!

Core Requirements

1. Personalized Student Profile System

- Input: Your student ID (hardcoded in the program)
- Output: Personalized profile showing your academic standing
- Implementation: Use your ID to generate unique academic patterns

2. Smart Daily Schedule Manager

- Display YOUR actual class schedule for the current day
- Show free time slots between classes
- Calculate time remaining until next class
- Suggest whether you should go to your hostel room, library or the canteen.

3. Academic Progress Tracker

- Calculate required SGPA to reach your target CGPA
- Track credit completion progress
- Visualize progress toward graduation
- Implement the required SGPA calculation using Plaksha's official CGPA/SGPA calculation rules (use your current CGPA, completed credits, current semester credits, and target CGPA accordingly)

4. Attendance Planning System

- For each subject, calculate classes needed to reach attendance cutoff
- Use university's/course official attendance policy to determine how many additional classes are required to meet the attendance percentage cutoff
- Show current attendance status and alerts

5. Activity Recommendation Engine

- Suggest study sessions, games, or club activities during free slots
- Recommendations should be personalized based on your schedule gaps
- Consider time of day and duration for appropriate suggestions

Personalization System

Your program MUST be personalized using the following scheme:

Base Data (You MUST customize these):

Base Data Template

```
student_id:        .word YOUR_ACTUAL_ID
current_cgpa:      .float YOUR_CURRENT_CGPA
target_cgpa:       .float YOUR_TARGET_CGPA
total_credits:     .word YOUR_COMPLETED_CREDITS
semester_credits:  .word YOUR_CURRENT_SEM_CREDITS
```

Personalization Algorithm

Your schedule, course preferences, and activity suggestions will be derived from your student ID using bit manipulation and modular arithmetic:

Example Personalization Techniques

```
# Example personalization techniques you should implement:
# - Course distribution pattern: (ID % 8) determines class timing template
# - Activity preferences: (ID >> 2) % 4 determines your "campus personality"
# - Study patterns: Sum of ID digits modulo 5 determines optimal study times
```

These are examples; you should design and document your own consistent personalization logic.

Expected Output Format

Your program should display a comprehensive dashboard:

Sample Output Layout

```
=== CAMPUS BUDDY ===
Student: [Your ID] | [Current Day]

 TODAY'S SCHEDULE:
[Time] [Activity/Class] [Location]
[Free slots with suggestions]

 ACADEMIC OVERVIEW:
Current CGPA: X.XX | Credits: XXX/XXX
Target CGPA: X.XX | Required this semester: X.XX

 ATTENDANCE PLANNER:
[Subject]: X/X classes (X%) → Need Y of next Z classes

 SMART SUGGESTIONS:
[Personalized activity recommendations]

Menu: [1-Schedule] [2-Grades] [3-Attendance] [4-Suggestions] [5-Exit]
```

You do *not* need to implement interactive input handling for the menu; a well-structured, readable summary output is sufficient.

Submission Requirements

1. Source Code File

- `campus_buddy_[StudentID].asm`
- Well-commented code explaining key algorithms
- Proper MIPS assembly structure with `.data` and `.text` sections

2. Demo Report (PDF)

- Student Information: Name, ID, Program
- Personalization Explanation: How your ID affects your program's behavior
- Feature Demonstration: Screenshots/output of key features working

3. Testing Instructions

- Brief instructions on how to test your specific features
- Any special inputs needed to demonstrate personalization

Grading Rubric

Category	Weight	Criteria
Personalization	5 Marks	Effective use of student ID for unique behavior
Functionality	8 Marks	All core features implemented and working
Code Quality	4 Marks	Clean, well-commented, efficient MIPS code
User Interface	2 Marks	Clear, organized output and easy navigation
Documentation	1 Marks	Clear demo report and testing instructions

Important Policies

- **Individual Work:** This is an individual assignment. While discussion is encouraged, code copying will result in zero marks.
- **AI Assistance:** You may use AI for learning concepts, but the final code must be your own.
- **Final Date and Time of Submission:** 25 November 2025, 11:59 PM.
- **Late Submission:** A penalty of 1 mark per day for the first day late, and 2 marks per day for each subsequent day.
- **Testing:** We will run your code with YOUR student ID to verify personalization works correctly.

Good luck, and enjoy creating your very own **OP-CODE, your campus buddy** !!